

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2023-24

Paper: BVCA-305 — Network Configuration and Management

Time: Three hours

Maximum Marks: 70

Instructions: Answer all sections. Candidates are required to write their answers in their own words as far as practicable. Symbols have their usual meanings.

SECTION — A

Long Answer questions. Answer any two questions:

2 × 12 = 24 Marks

Question 1. What is a Layer 3 switch? Explain the process of inter-VLAN network communication with a suitable diagram.

Question 2. Describe the various layers of the OSI model with a suitable diagram.

Question 3. What is WLAN? Explain the components of WLAN in detail.

Question 4. Consider the IP address 8.1.4.5/16. Find:

- (a) the number of host bits,
- (b) the number of subnet bits,
- (c) the number of subnets,
- (d) the number of hosts per subnet.

Assume only one subnet mask is used throughout the network.

SECTION — B

Question 5. Attempt any six parts (Answer in about 200 words each): **6 × 6 = 36 Marks**

- (a) What are the various network devices? Explain their working.
 - (b) Describe and compare Hub, Bridge, and Switch.
 - (c) Discuss the Ethernet CSMA/CD protocol.
 - (d) What do you understand by CAPWAP protocol?
 - (e) What is WLC? What are its various deployment methods?
 - (f) Explain the SFTP protocol in network management.
 - (g) What is MAC addressing? How does it differ from IP addressing?
 - (h) How is PRTG helpful in network monitoring and management?
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- (i) Describe the functionality of a Router and compare it with a Switch.

SECTION — C

Question 6. Attempt all parts:

10 × 1 = 10 Marks

- (a) SNMP stands for _____.
- (b) In the OSI model, network cables are considered to function at Layer _____.
- (c) In network models, routers are considered to operate at Layer _____.
- (d) The WLC and APs use the _____ protocol to communicate.
- (e) The new IP addressing schema (128-bit) is known as _____.
- (f) For a network with a CIDR prefix of /16, its dotted decimal equivalent is _____.
- (g) The equivalent of subnetting at Layer 2 is _____.
- (h) Is UTM a firewall or not?
- (i) Which protocol is more reliable for data transmission — TCP or UDP?
- (j) Which command/utility is used to trace the routing path of a packet across a network?